

Experiment 10

Problem Statement

You are a DevOps engineer responsible for ensuring the reliability, availability, and observability of a deployed web application.

Design and implement a monitoring and incident management system using free tools such as Prometheus, Grafana, and Alertmanager, deployed via Docker Compose.

Tasks

Task 1: Develop and Containerize the Application

1. Create a simple web application (Node.js or Python Flask) with:
 - “/” endpoint returning a success message
 - “/error” endpoint that generates an intentional failure
 2. Containerize the application using Docker.
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Task 2: Set Up Monitoring Infrastructure

1. Create a docker-compose.yml file to deploy the following services:
 - Web application
 - Prometheus
 - Grafana
 - Node Exporter

2. Configure Prometheus to:
 - Scrape system metrics from Node Exporter
 - Scrape application metrics (if available)
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Task 3: Configure Visualization Dashboard

1. Access Grafana through a web browser.
 2. Add Prometheus as a data source.
 3. Create dashboards to display:
 - CPU usage
 - Memory usage
 - Request rate
 - Error rate
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Task 4: Define Reliability Metrics

1. Define and calculate the following:
 - Service Level Indicator (SLI)
 - Service Level Objective (SLO)
 - Availability percentage
 - Error rate
2. Clearly document how each metric is computed.

Task 5: Configure Alerts

1. Define alerting rules in Prometheus for:
 - High CPU usage
 - Application downtime
 - High error rate
2. Configure Alertmanager to:
 - Receive alerts from Prometheus
 - Send notifications via email or webhook

Task 6: Simulate Incidents

Perform the following failure scenarios:

1. Stop the application container
2. Continuously invoke the “/error” endpoint
3. Simulate high CPU usage

Observe and record:

- Changes in metrics
- Triggered alerts

Task 7: Perform Incident Management

1. Detect the incident using dashboards and alerts
 2. Diagnose the root cause of failure
 3. Perform recovery actions such as:
 - Restarting the container
 - Scaling the application
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Task 8: Improve System Reliability

1. Implement:
 - Health checks
 - Restart policies
2. Scale the application to multiple instances

Submission Instructions:

1. Students must upload a single PDF file containing the report of the experiment conducted.
2. The report should contain docker-compose.yml configuration file, screenshots of Grafana dashboards, alert configuration files, incident report including: description of incident, detection method, root cause analysis, resolution steps, preventive measures.
3. The PDF name should be in the form "StudentName_SAPID".